Fedy Ben Hassouna

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Skills

Languages: Python, SQL, C, C++, Bash, PowerShell
Machine Learning and Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, HuggingFace Transformers
Data and NLP Tools: Pandas, OpenCV, NLTK, SpaCy, LangChain, LangFlow, LangGraph, ChromaDB, FAISS
Web and APIs: Flask, FastAPI, REST APIs, Tkinter, Streamlit
LLMs and Vector Search: LLaMA, GPT, RAG pipelines, Semantic Search, Neo4j, Knowledge Graphs, Milvus
Agentic AI Systems: CrewAI, Agno, AutoGPT, multi-agent orchestration
GenAI Applications: Prompt Engineering, Function Calling, Toolformer, Vision-Language Models (VLMs)
DevOps and Automation: Git, GitHub Actions, Docker, Firebase, n8n, CI/CD workflows
Other: BeautifulSoup, Whisper ASR, Segment Anything (SAM), GitHub Copilot, OCR, Tesseract
Spoken Languages: Arabic (Native), English (Conversational), French (Conversational)

Experience

AI Engineer, Part time ,WiseVision AI technologies – Tunis February 2025 – June 2025

- **AI Agent Development:** Led the design of an AI agent for a major Tunisian retailer, capable of autonomously analyzing complex datasets, generating visualizations, and delivering actionable business insights to support strategic decisions.
- **BI Automation:** Built AI-driven tools to automate business intelligence from raw sales and operational data, enhancing decision-making and boosting market responsiveness.

Embedded AI and Robotics Developer , Part time ,Junior Robotics – Tunis September 2024 – March 2025

- **Robot Development:** Built an autonomous robot equipped with a camera and a robotic arm to identify and collect trash, delivering it to a designated trash can automatically. Focused on integrating AI and computer vision for real-time object detection and robotic control.
- **Practical Workshops:** Led interactive workshops to guide students through building and programming their own autonomous robots, focusing on automation, AI, and hardware integration.

MLops Engineer, Internship ,SeekMake – Tunis September 2024 – October 2024

- **Cloud Integration:** Enhanced the prompt during deployment and integrated AI models on Azure AI Studio, aligning them with the current billing framework and avoiding external payment setups.
- **Billing and Resource Management:** Have compared two models, GPT-4o and Mixtral, to manage the billing framework within Azure, ensuring operational efficiency and cost-effectiveness.

Computer Vision Engineer , Internship , MUST University – Tunis June 2024 – August 2024

- **Designed** a deep learning model for detecting cork oak trees using aerial imagery and segmentation techniques.
- **Optimized** detection with Mask R-CNN architecture, leading to improved forest management by accurately estimating biomass.

Education

INSAT : National Institute of Applied Sciences and Technology
Master Degree in Automation, Embedded Software, and Industrial IT Expected Graduation: 2026

Achievements

1st Place at Hack for Good 3 – Pump Watch: AI-Powered Predictive Maintenance

- Developed an AI-driven predictive maintenance solution for industrial pump stations, utilizing real-time monitoring and frequency domain analysis of motor currents.
- Integrated "Pumpo," an intelligent chatbot providing instant maintenance insights, diagnostic assistance, and scheduling support.
- Represented Enactus INSAT and secured 1st place at the Hack for Good 3 Hackathon organized by Junior Entreprise INSAT.

1st Place in the TN 2056 ICC TSYP12 Challenge – AyyusLabs

- Developed an innovative solution as part of a 6-month hackathon, securing 1st place in the TN 2056 ICC TSYP12 Challenge.
- Represented the National Institute of Applied Science and Technology (INSAT), showcasing innovation and excellence in AI-driven solutions.

1st place at the HACK-E Hackathon with DHAWWINI – Tunis Sciences City

- DHAWWINI: An innovative energy solution combining a mobile solar panel and an AI-controlled turbine to meet the energy needs of unelectrified urban areas, offering a sustainable and efficient approach .

Accepted into the Ready Tensor, Inc. Computer Vision Projects Expo 2024 | Real-Time Protest Banner Analysis Using AI

- Developed an AI-driven system that interprets text from protest banners in real-time, providing valuable insights for social and political research.

Projects

Voice-Controlled Robot using Whisper - GitHub

- Built a real-time voice-command system using OpenAI's Whisper model to control a mobile robot via natural language.
- Enabled seamless speech-to-action execution with fast audio preprocessing and response generation, showcasing robust edge AI capabilities.

Excusely: AI Workflow Automation with n8n and Firebase - GitHub

- Developed a modular automation pipeline using n8n, Firebase, and OpenAI API to generate and manage user-specific excuses.
- Demonstrated practical deployment of AI using low-code tools, serverless functions, and prompt-driven generation.

Knowledge Graph-Enhanced RAG System - GitHub

- Engineered a hybrid system that integrates a knowledge graph with Retrieval-Augmented Generation (RAG) to improve LLM reliability.
- Leveraged LangChain and Neo4j to enable contextual, grounded answers in generative applications such as enterprise chatbots.